

Comparative and meta-analytic insights into life extension via dietary restriction.

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Dietary restriction (DR) extends the lifespan of a wide range of species, although the universality of this effect has never been quantitatively examined. Here, we report the first comprehensive comparative meta-analysis of DR across studies and species. Overall, DR significantly increased lifespan, but this effect is modulated by several factors. In general, DR has less effect in extending lifespan in males and also in non-model organisms. Surprisingly, the proportion of protein intake was more important for life extension via DR than the degree of caloric restriction. Furthermore, we show that reduction in both age-dependent and age-independent mortality rates drives life extension by DR among the well-studied laboratory model species (yeast, nematode worms, fruit flies and rodents). Our results suggest that convergent adaptation to laboratory conditions better explains the observed DR-longevity relationship than evolutionary conservation although alternative explanations are possible. © 2012 The Authors. *Aging Cell* © 2012 Blackwell Publishing Ltd/Anatomical Society of Great Britain and Ireland.